

Daffodil International University

Department of Software Engineering,

B.Sc in SWE

Syllabus

SE 111 Computer Fundamentals

Basic computer system, computer codes, convert number systems, logic gates, Application of logic gate, Basic concepts on microprocessors and microcomputers. Data and information, variables, Loops, Condition, identify errors in logical flow in flowcharts, correct the logic of the control structures of simple programs in programming language. Various parts of a computer system, input/output devices, Memory hierarchy, Types of memory, Memory operation, Data communication, Types of networks.

SE 112 Computer Fundamentals Lab

The Computer Fundamentals Lab is designed to provide students with a hands-on understanding of core concepts in computing, programming, and essential software tools. Throughout this course, students will delve into the fundamentals of data, programming logic, and software applications, enabling them to build a strong foundation in computer science and information technology. Emphasis will be placed on practical implementation, ensuring students gain real-world skills that are applicable in various academic and professional settings.

SE 113 Introduction to Software Engineering

Software engineering fundamentals, software process, software process models, methodologies; prototyping, iterative process models, incremental software development, agile software development, extreme programming, Kanban, and SCRUM; Software requirement Engineering: functional and non-functional requirement, requirement engineering process, requirement elicitation, specification, validation, and change; software design. software modeling, UML diagrams; software testing, different types of testing, test-case design, white box testing: basis path testing with cyclomatic complexity, black box testing: interface testing, equivalence partitioning, boundary value analysis, object-oriented software testing: class testing, behavioral testing, Halstead's complexity, Decision table; maintenance, documentation, reliability engineering, software quality and security, reverse engineering, software risk management, software project management: phases of software project management, estimation techniques, scheduling techniques, COCOMO models.

ENG 101 English I

Overview of Basic Skills of English: Speaking, Basic Grammar: Use of Articles, Identifying Parts of Speech, Syntax: Run on, Fragments, speaking: Self Introduction, Social English: Greeting, Answering to Greeting, Agreeing & Disagreeing, Small Talks etc., Relating and Demonstrating Ideas on Selective Topics; Introduction to IELTS Speaking: Part I and II, Reading: Reading techniques (Skimming and Scanning), Reading Comprehension: Practices from IELTS Reading Comprehension: (True/False/Not Given, Flow chart, Matching Heading, Matching Features, Multiple Choice Question, Short Answer etc.), Writing: Formal Letter Writing, Cover Letter, CV, Resume, Video Resume, E-mail Writing, Paragraph Writing, Essay Writing.

BNS 101 Bangladesh Studies

Geography location, borders, topography, resources, climate; Demographic traits: demographic dividend, human development; Society and culture: Social stratification, traditions, values, festivals; Ethnic identity of people of Bangladesh; Origin and development of the name of Bangladesh; Origin and development of Bangla language; Bangladesh in international affairs: principles, objectives and determinants of foreign policy, achievements and challenges; Constitution of Bangladesh: concept, essentials, principles, rights, and amendments; Organs and functions of government: Digitalization and Service Sector in Bangladesh; Historical background of Bangladesh: Language movement, Six-points demands, Liberation war, constitutional development in post -liberation; Economic profile of Bangladesh: macro perspective, blue economy, gig economy; Development approaches: Vision 2041, SDGs, Delta plan; Agricultural Productivity and Rural development; Urbanization: push-pull model of migration; Environmental degradation and climate change; Industrial sector: overview of major industries, ICT & Software sector of Bangladesh.

MAT101 Mathematics-I

Function, Domain, Range, Graph, Limit, Continuity & Differentiability, Derivatives; Differentiation and Successive Differentiation of various types of function; Leibnitz's Theorem, Roll's Theorem; Mean value Theorem; Taylor's and McLaurin's Theorem in finite and infinite forms, Euler's Theorem; Maximum and minimum values of functions of single variable, Expansion of functions; Evaluation of indeterminate forms; Partial differentiation, Integration by the method of substitutions, successive reduction; Integration by parts; Standard integrals; Definite integrals: properties and use in summing series; Walli's Formula, Improper integrals, beta function and gamma function; Jacobian; Multiple integral and its application, Area and volume between curves and axes.

SE 121 Structured Programming

Overview of C: Basic structure of C programs, Importance of learning C programming, Execution of a C program. Constant, Variables, and Data Types: Different data types and their specifiers, declaration of variables, Assigning values to variables, taking input using scanf() function, Operators and Expressions: Arithmetic operator, Relational operator, Logical operator, Increment

Decrement operator, Conditional operator, Conditional Statements: simple if statement, if...else statement, Nesting if... else statement, if...else if ladder, switch statement. Loops: while loop, do..while loop, for loop, nested for loop, continue and break, Array: Declaration and Initialization of 1D and 2D arrays. Strings: Declaration and Initialization of String variables, arithmetic operation on characters, string handling functions, User-defined Functions: Definition of user-defined function, Elements of user-defined functions, Category of user-defined functions, Structures: Defining a structure, declaring structure variables, Accessing structure members, Structure initialization.

SE 122 Structured Programming Lab

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SE 123 Discrete Mathematics

The syllabus of the course has been divided into four core sections. These are propositional & predicate logic, set theory & probability, function & relation, and graph theory & binary tree. These four sections have been expanded with necessary subsections. The propositional & predicate logic starts with a simple and compound logical statement. After that, with the discussion and application of logical connectives and the truth table, the students learn to apply the concept to solve relevant problems. The set theory and probability contain the basic terminologies, sample space formation, and probability concepts in appropriate contexts. The function and relation section introduces different types of functions, their scope of application, and the comparison between function and relation. Finally, graph theory and binary tree cover the terminologies of graph theory, different types of graphs, and the essentials of the binary tree.

SE 212 Software Requirements Specification & Analysis

Basic software requirements, Requirements Engineering, Functional and Non-Functional requirements, Feasibility Study, Stakeholders for a system, design user profile, Requirements Elicitation Techniques, Requirements Representation, SRS documentation, Use Case, Requirements Modeling, Actor of a use case, Include and Extended Use Case, Case description

Requirements Prioritization Techniques, Requirements Traceability Matrix, Prototyping, Usability and UX, SUS calculation.

SE 213 Digital Electronics & Logic Design

Foundation of Digital Electronics, Fundamental concepts, digital and analog quantities, advantages of digital electronics, Introduction to Number Systems, Conversion Number Systems, Expression, truth table, universal properties of NAND & NOR gates, Simplification of expression using Boolean algebra, Binary Subtraction using 1's and 2's complements, Design of Combinational Logic using TT, Boolean algebra, Adder Circuit, Half Adder, Full Adder, Parallel Adder, Introduction to K-Map, SOP, POS, MINTERM, MAXTERM, 3-4 variable K- MAP, construction of map, implementation of expression using map, grouping and simplification, Advanced Combinational Circuits, Decoder (Binary to Decimal, Octal, Hexadecimal, Advanced Combinational Circuits, Converters (BCD to XS- 3), Introduction to Multiplexers and Demultiplexer, Design of Multiplexer Circuits, Design of De- multiplexer Circuits, Introduction to Parity, Design of Parity Generator Circuits, Design of Parity Checker Circuit to detect error during transfer or transmission of Data, Introduction to comparator circuit using complex combinational Logic, Introduction to Computer Memory, Introduction to Flip Flops, Design of J-K, S-R, Latch.

PHY101 Physics I

Mechanics: Overview; history; classical mechanics; Newton's laws of motion; Motion in two dimensions: Projectile motion and its application; Force and friction force, Work & Energy, Momentum and Moment of inertia Waves and oscillations: Concepts of waves & oscillations; Simple harmonic motion: energy, composition, damped and forced vibration; types of waves, stationary and progressive wave and their applications. Optics: Concepts of the historical background of optics, classifications: Geometrical optics; reflection, refraction, total internal reflection; Physical optics: Interference, diffraction, polarization, their uses. Atom and Modern Physics: Atomic structure and atomic development, atomic nucleus and nuclear energy. Basic concepts of modern physics: Theory of relativity; Einstein's photoelectric effect, Compton effect, pair production, X-Ray. Radioactivity.

MAT 102 Mathematics II

Linear Algebra: Basic of Matrix and Matrix Algebra, Types of Matrix, Determinant of Matrix (Higher Order), Inverse Matrix, Rank of Matrix, RREF & NF of a Matrix, System of Linear Equations, Eigenvalues and Eigenvectors, Cayley-Hamilton Theorem; Complex Variables: Basic of Complex Numbers, Modulus, Argument, Different form of Complex Number, Cauchy Riemann Equation, Laplace's Equation, Harmonic Function, Euler Theorem, De Moivre's Theorem; Fourier Analysis: Fourier series, Fourier Transformation.

SE 131 Data Structure

Overview of Data Structures, Algorithm and Complexity, Time Space Tradeoff, Recursion, Iteration, Record, Pointer, Implementation of Memory, Array: Traverse, Insert, Insert at any position, Delete at any position, Linear Search, Linear Search complexity, Binary Search, Binary Search Complexity, Searching- Bubble Sort, Factorial and Tower of Hanoi Problem; Marge, Stack: Stack data Insertion, Stack Deletion, Stack : Search, Prefix, Infix and Postfix Expressions, Queue: Queue data Insertion, Queue Deletion, Queue: Search, Double Ended Queue, Priority Queue, Hashing: Hash Indices and Hash Functions, Static Linked List: List ADTs, Linked List data Insertion, Deletion, Search, Double Way Linked List: data Insertion, Deletion, Search, Circular Way Linked List: data Insertion, Deletion, Search, Tree, Tree terminologies, Binary Search Tree: Insertion, Deletion, Search, AVL Tree, Heaps, Heap Sort, Graph, Graph Terminologies, Adjacency matrix, Adjacency List, Graph data Insertion, Graph Deletion, Graph: Search.

SE 132 Data Structure Lab

Array – Traverse, Insert, Insert at any position, Delete at any position, Linear Search, Linear Search complexity, Binary Search, Searching- Bubble Sort, Stack – Stack data Insertion, Stack Deletion, Stack – Search, Prefix, Infix and Postfix Expressions, Queue – Queue data Insertion, Queue Deletion, Queue – Search, Double Ended Queue, Priority Queue, Static Linked List – List ADTs, Linked List data Insertion, Linked List Deletion, Linked List – Search, Double Way Linked List – Double Way Linked List data Insertion, Double Way Linked List Deletion, Double Way Linked List – Search, Circular Way Linked List – Circular Way Linked List data Insertion, Circular Way Linked List Deletion, Circular Way Linked List – Search, Tree, Binary Search Tree – Tree data Insertion, Tree Deletion, Tree – Search, AVL Tree, Heaps, Heap Sort, Graph, Graph Terminologies, Adjacency matrix, Adjacency List, Graph data Insertion, Graph Deletion, Graph – Search.

SE 133 Software Development Capstone Project

The “Software Development Capstone Project” course provides hands-on experience and practical application of key programming concepts using structured programming language. Students will be guided through a systematic approach to develop a project, emphasizing problem-solving skills and logical thinking. The course covers a range of topics from problem formulation to the evaluation of solutions, ensuring that students acquire a comprehensive understanding of the software development process. The course encompasses fundamental programming elements, placing particular emphasis on data types, variables, condition statements, loop control structures, switch cases, functions, arrays, pointers, file operations, and string manipulation. These core concepts form the backbone of the curriculum, providing students with a comprehensive understanding of essential programming principles.

SE 216 Object Oriented Programming

Basic Concepts of Object-Oriented Programming, Classes and Objects, Characteristics of Objects, Relationships and Methods, Advantages of OOP over structured programming, Access specifiers, Static and non-static members, Constructors and Its types, Destructors, Array of objects, object pointers, and object references, Abstraction, Encapsulation, Inheritance, Polymorphism, Abstract classes and methods, Abstract methods, Abstract layer, Basic principles of Interfaces, Extending Interfaces, Multiple Interfaces, Interface Implementation, Definition of Refactoring, Different Types of Code Smells, Refactoring Techniques, Functions and Modules, Library like List, set, stack, queue, Tuple, Dictionary etc., Exception Handling, File Handling, Software Engineering Life Cycle, Modeling Problem and Solution, Moving to Code, Internal Component Definition, Design for Reuse, Design Class Diagrams, Iterating the Design, Refactoring, Design Best Practices, Cohesion, Complexity, Coupling, Congruence, Singleton pattern, Factory pattern, Observer pattern, Strategy pattern, Use Case Model, Use Case Diagram, Use Case Description, Main Success Scenario and Alternate Paths, Use Case Relationships (Generalization, <<extends>>, <<includes>>), Extension Points and Packages, Overview and Basic Concepts, Classifiers and Well-Formedness Rules, Basic Notation, Classes, Objects, Relationships, and Methods, UML Diagrams.

SE 217 Object Oriented Programming Lab

Setting Up the Development Environment, Introduction to Classes and Objects, Implementing Access Specifiers, Working with Static and Non-Static Members, Constructor and Destructor Implementation, Handling Arrays of Objects and Object Pointers, Implementing Abstraction, Encapsulation in Action, Working with Inheritance, Exploring Polymorphism, Abstract Classes and Methods Implementation, Interface Implementation, Applying Refactoring Techniques, Creating and Using Libraries, Exception Handling in Practice, File Handling, Software Engineering Life Cycle, Transition from Design to Code, Design Class Diagrams, Refining Design Practices, Assessing Cohesion, Complexity, and Coupling, Implementing Design Patterns, Creating Use Case Diagrams, Use Case Descriptions and Relationships, UML Diagrams in Action.

SE 222 Computer Architecture

Instruction Set Architecture, Evolution of Computer, Processor performance (Moore's law), Benchmarks (Ahmdahl's law), clock speed and instruction per second, General Purpose Processor and its instruction patterns, Function of GPP, Hypothetical Processor, Instruction fetch and execute, Logical Units, Instruction Cycle States, Interrupt and Interrupt handling, Interconnection structure, Bus interconnection, Bus design (Address bus, control bus, data bus, multiple bus), Traditional bus architecture, high performance architecture, Elements of bus design, Method of arbitration, Timing, performance of memory, The memory hierarchy, Cache memory principles, Cache address, Mapping mechanism, Direct mapping, Full and Set associative cache, Replacement algorithms, Write policy, Line size, Number of cache, Interconnection structure, Data flow, Arithmetic logic unit (ALU), Number representation, Addition subtraction technique, Multiplication Technique (1 and 2's complement), Floating point representations, Machine

instructions characteristics, Elements of machine instruction, Instruction representation, Instruction type, Number of address, Instruction set design, Operand types, Operation types, Branch instructions, Skip instructions, Procedure call instructions, Addressing technique, CPU internal structure, Register Organization, User visible registers (flags), Condition codes, Control and Status registers, Data flow, Instruction pipelining, Pipeline performance, Pipeline hazards, Dealing with branches.

STA 101 Statistics I

Meaning and Definition of Statistics, Types of statistics, Population and sample, Parameter and statistic, Variable and types of variable; Characteristics Levels of data, Sampling techniques, Define data presentation, Constructing frequency distribution and relative frequency, percentage frequency and cumulative frequency distribution both for Qualitative and quantitative data; Graphic presentation of a quantitative frequency distribution (Histogram, Frequency polygon, Ogive curve) with their merits and demerits; Graphic presentation of a qualitative frequency distribution (Bar Chart, Pie Chart) with their merits and demerits, Time series data, Line diagram; Define Measures of Central Tendency, applications, Arithmetic Mean, Geometric Mean, Harmonic Mean, Weighted Mean for ungroup data and grouped data with their merits and demerits; Calculation of Median, mode both for group and ungroup data, Application of measures of central tendency for different types of level of measurements. (Definition of Measures of Location, concepts of Quartiles, Percentiles, and Deciles, Procedure of calculating quartiles, deciles, and percentiles for ungrouped data. Measures of Location (Procedures of calculating Quartiles, Percentiles, and Deciles for group data), mathematical problem solving with interpretations. Calculation of IQR with interpretation, Drawing Box and Whisker Plot with its applications. Define measures of dispersion, applications, types of dispersion, Absolute measures, Relative Measures, and Calculation of Mean deviation. Calculation of Population variance, Population Standard Deviation, Sample variance, Sample Standard Deviation, and Coefficient of Variation for un group data with mathematical problem-solving. Calculation of Population variance, Population Standard Deviation, Sample variance, Sample Standard Deviation, and Coefficient of Variation for group data with mathematical problem-solving. Define the shape of the distribution, Concepts of Skewness and Kurtosis with graphical presentation. Calculation of Coefficient of skewness and coefficient of Kurtosis with their interpretations. Define correlation Analysis, Types of Correlation, positive correlation, negative correlation, simple correlation, partial correlation, multiple correlation, linear correlation, non-linear correlation, Scatter diagram with interpretation. Calculation of Coefficient of correlation with interpretation, mathematical problem-solving. Define Regression Analysis, Types of Regression, Simple regression analysis, Multiple regression analysis, Calculation of Simple regression coefficients. Interpretation of regression coefficients, Calculation of the standard error of the regression coefficients, and Calculation of coefficient of determination with interpretation. Define Probability, equally likely outcomes, mutually exclusive outcomes, Sample Space, Tree diagram, Venn diagram, Laws of probability, Additional rules. Marginal probability, Joint probability, Conditional probability, multiplication rules complement rule. Bayes theorem with related math. Random variable, Discrete Random variable, Continuous random variable, Mean and variance and standard deviation of random variables. Random variable, Discrete Random variable, Continuous random

variable, Mean and variance and standard deviation of random variables. Define Probability distribution, Basic idea of Discrete Probability Distribution/ Probability Mass Function (PMF) and Continuous Probability Distribution/ Probability Density Function (PDF). Probability Distribution (Bernoulli distribution and Binomial Distribution with related maths, Geometric and Poisson Probability distribution with related math, Normal and exponential probability distribution with related math). Basic concepts of Test of hypothesis, Null hypothesis, Alternative hypothesis, Test Statistics, One-tailed test, Two-tailed test. Basic idea of One sample test and two sample test, Z-test, t-test, critical value, Acceptance area, rejection area, Decision rule. One sample Z-test and t-Test with related math and interpretation. Two sample Z-test and t-Test with related math and interpretation

AOL 101 Art of Living

It has been perceived from different evaluation that intellectual ability needs to be exercised in an effective way through scheduled practice and performance. Success comes not just from the knowledge and skills but also from behavior which can create distance between success and failure. And we can achieve expected behavior only by learning and unlearning ways of thinking that finally craft body language and ensure effective living in this beautiful world. This course assists the students to achieve the ability to understand life positively in personal, family and professional levels.

SE 211 Object Oriented Concepts

The four pillars of object-oriented programming – Abstraction, Encapsulation, Inheritance, and Polymorphism. Abstract class and interface. Definition of Refactoring and Code Smells. Different types of code smells and refactoring techniques. Use of different refactoring techniques for different purposes. Introduction to Design Patterns. Definition and classification of design patterns. Definition, Use Case, Code examples of Creational Patterns – Singleton, Factory Method, Abstract Factory, Prototype, Builder. Definition, Use Case, Code examples of Creational Patterns – Adapter Pattern, Bridge Pattern, Composite Pattern, Decorator Pattern, Facade Pattern, Flyweight Pattern, Proxy Pattern. Definition, Use Case, Code examples of Creational Patterns – Chain of Responsibility Pattern, Command pattern, Interpreter pattern, Iterator pattern, Mediator pattern, Memento Pattern, Observer Pattern, State pattern, Strategy Pattern, Template pattern, Visitor pattern.

SE 221 Object Oriented Design

The Object-Oriented Paradigm (Basic Concepts Classes and Objects, Characteristic of Objects, Relationships Methods), Object-oriented programming (Programming as a tool, More than one programming as a language, Data-types, Variables, Basic operators, Basic I/O, Boolean values, Conditional execution loops, List and list processing, Logical and bitwise operations, functions, tuples, dictionaries and data processing, modules, packages, string, list methods, exceptions), Object oriented approach (Classes, Methods, Objects, exception handling, working with files.), Analysis, Design, Modeling (Systems and Software Quality Software Engineering Life Cycle

Properties, Analysis and Design, Structured vs. Object Approach Modeling, Modeling Problem and Solution, Modeling Perspective), Processes and Approaches to OOA&D (Unified Process, Rational Unified Process, Agile Approaches, Other Approaches), Design Phase (Moving to Code, Internal Component Definition, Design for Reuse, Design Class Diagrams, Operations, Methods), Designing Quality into Modules (Iterating the Design, Refactoring, Design Best Practices, Cohesion, Complexity, Coupling, Congruence), Physical Design, (UML Implementation Diagrams, Component Diagram, Application of the Component Diagram, Component Diagram Notation and Syntax, Deployment diagram, Application of the Deployment Diagram, Deployment Diagram Notation and Syntax), Domain Modeling (The Information Model (Structural Modeling, Objects, Attributes, and Associations, Links and Messages, Information Modeling in the UML, Basic Constructs, Relationships, Stereotypes, Packages), Use Case and the Behavior Model (Use Case Model, Use Cases in the Life Cycle, Use Case Diagram, Use Case Description, Main Success Scenario, Alternate Paths, Preconditions and Post- Conditions, Exception Paths, Use Case Extensions, Use Case Relationships, Generalization, The <<extends>> Relationship, The <<includes>> Relationship, Extension Points, Packages), Basic Unified Modeling Language (Overview, Basic Concepts, Classifiers, Well-Formedness Rules, Basic Notation, Classes and Objects, Characteristic of Objects, Relationships and Methods, Other Notation, UML Diagrams)

SE 214 Algorithm Design & Analysis

Basic strategies of algorithm design: top-down design, divide and conquer, average and worst-case criteria, asymptotic costs. Simple recurrence relations for asymptotic costs. Choice of appropriate data structures: arrays, lists, stacks, queues, trees, heaps, priority queues, graphs. Applications to sorting and searching, matrix algorithms, shortest-path and spanning tree problems. Introduction to discrete optimization algorithms: dynamic programming, greedy algorithms. Graph algorithms: depth-first and breadth-first search.

SE 215 Algorithm Design and Analysis Lab

Big O Notation, Space and time complexity. Function and Recursion. Choice of appropriate data structures: Arrays, Lists, Stacks, Queues, Trees, Heaps, Priority Queues, Graphs. Sorting algorithms: Bubble Sort, Improved Bubble Sort, Selection Sort, Merge Sort, Quick Sort, Radix Sort. Search algorithms: Linear Search, Binary Search. Matrix Algorithms. Introduction to discrete optimization algorithms: dynamic programming techniques, Fractional Number, Knapsack, Longest Common Subsequence, Greedy algorithms: Coin Change Problem, Huffman Coding. Graph representations: Adjacency List, Adjacency Matrix. Graph algorithms: Depth First Search, Breadth First Search, Shortest Path, Minimum Spanning Tree, Floyd Warshall, Dijkstra Algorithm, Bellman Ford, Kruskal's Algorithm, Prim's Algorithm.

SE 235 Desktop & Web Programming

The Desktop & Web Programming course offers a comprehensive exploration of software development, covering essential concepts and practical skills necessary for crafting applications

across desktop and web platforms. Encompassing programming fundamentals, frontend and backend development techniques, application security, and project deployment strategies, this course equips students with a robust understanding of creating functional, user-centric, and secure software solutions in both desktop and web environments. Additionally, the course touches upon emerging trends, providing insights into the future landscape of software development.

SE 236 Desktop & Web Programming Lab

Setting up the development environment for Java, Basic Java syntax, and programming concepts, Hands-on exercises on variables, data types, and operators, Control structures in Java (loops and conditionals), Object-Oriented Programming (OOP) principles in Java. Implementing classes, objects, inheritance, and polymorphism in Java, Creating a basic HTML5 webpage, Introduction to CSS3 and styling web content, Building basic web pages using HTML and CSS, Fundamentals of JavaScript language, Manipulating the Document Object Model (DOM). Basic JavaScript functions and events handling, Introduction to a frontend framework (React, Angular, or Vue.js), Hands-on development using the selected framework, Implementing state management and component-based architecture, Server-side scripting with Node.js, Python, or PHP, Integrating databases (SQL, NoSQL) in backend development, Creating RESTful APIs for web applications, Identifying and mitigating common security threats, Implementation of security measures (authentication, encryption), Hands-on exercises on secure coding practices and avoiding vulnerabilities, Introduction to desktop application frameworks (JavaFX, .NET), GUI design principles and local data management, Project management, version control, and collaborative development practices

SE 223 Database System

Databases and Database Users: Introduction - An Example, Characteristics of the Database Approach, Advantages of Using the DBMS Approach. Database System Concepts and Architecture: Data Models, Schemas and Instances. Three- Schema Architecture and Data Independence, Database Languages and Interfaces, The Database System Environment, Centralized and Client/Server Architectures for DBMSs. Entity-Relationship (ER) Model :Using High-Level Conceptual Data Models for Database Design, An Example Database Application, Entity Types, Entity Sets, Attributes, and Keys - Relationship Types, Relationship Sets, Roles, and Structural Constraints, Weak Entity Types, Refining the ER Design for the COMPANY Database - ER Diagrams, Naming Conventions, and Design Issues The Relational Algebra and Relational Calculus: Unary Relational Operations: SELECT and PROJECT, Relational Algebra Operations from Set Theory. SQL Data Definition and Data Types, Specifying Constraints in SQL, Schema Change Statements in SQL, Basic Queries in SQL INSERT, DELETE, and UPDATE Statements in SQL, Views (Virtual Tables) in SQL. Functional Dependencies and Normalization for Relational Databases: Informal Design Guidelines for Relation Schemas, Functional Dependencies, Normal Forms Based on Primary Keys - General Definitions of Second and Third Normal Forms, Boyce-Codd Normal Form. Introduction to Transaction Processing, Transaction and System Concepts, Desirable Properties of Transactions, Characterizing Schedules

Based on Recoverability, Characterizing Schedules Based on Serializability, Concurrency Control Techniques: Two-Phase Locking Techniques for Concurrency Control.

SE 224 Database Systems Lab

The "Database Systems Lab" course provides students with a comprehensive exploration of fundamental introduction to DBMS, Purpose of DBMS, Relational Model, Data Abstraction, Database Architecture, Keys, ER Model, ER Model Scenario and ER Model, Relational Schema, ER Model to Schema Practice Example, Relational Algebra, Basic SQL Queries, Join Operation, More SQL Queries, Sub-queries in DBMS, Different Operation on Table, Stored Procedure and Views, Transaction, More about Transaction, Trigger, Normalization, Join Operation etc.

SE 232 Operating System and System Programming

This course introduces modern operating systems. It focuses on UNIX-based operating systems, though alternative operating systems, including Windows, are introduced. This course is beginning with an overview of the structure of modern operating systems. Throughout the subsequent units, discuss the history of modern computers, analyze in detail each of the major components of an operating system (from processes to threads), and explore more advanced topics in the field, including concurrency (synchronization, mutual exclusion, deadlock, starvation), memory (both primary and secondary) management and input/output file management and organization (segmentation, paging, swapping), file systems, and operating system support for distributed systems. Different CPU scheduling algorithms and disk scheduling algorithms have also been discussed in detail.

SE 233 Operating System & System Programming Lab

This course introduces the tools used to develop modern operating systems (OS). The focus of this course is UNIX-based operating systems, though alternative operating systems, including MS Windows are introduced. This course is divided into two parts. The first part is about the GNU/Linux command line interface to operate basic operations. The second parts are about the GNU/Linux application programming interface. In this part the creation and control of processes and threads are practiced. The cooperative process resource share management system and some advanced topics in the field of process scheduling, concurrency (synchronization, mutual exclusion, deadlock, and starvation), memory (both primary and secondary) management and input/output file organization are practiced.

GE 235 Principles of Accounting, Business and Economics

Clear Concept of Accounting in action, Accounting Equation and effects of Transaction Analysis and preparing the Financial Statement, Recording Process (Journal, Ledger and Trial Balance), Adjusting the accounts, Completing the accounting cycle with worksheet, case study, study on Financial report of a company, Foundation of Business, Barter Theory, Factors and Resources of Business, Proprietorship, Partnership and Corporate Business, Basic concept of Economics,

Micro and Macro Economics, Production Possibility Frontier, Factors of Production, Statements, Demand and Supply Curve with the schedule, Factor affecting on Demand and Supply, Law of Demand, Movement of the curve, Market Equilibrium.

SE 532 Introduction to Robotics

This course includes definition of Robot, Types of Robots (manipulator, legged robot, wheeled robot, autonomous underwater vehicles), Use of Robots, Asimov's laws of Robotics, History of Robotics, Key components of Robot, Sensors: Introduction, working principles and use of sensors (vision, force, LDR, temperature, smoke, accelerometer gyroscope, laser, tilt, compass, PIR, Infrared, etc.), Actuators and different actuators (DC motor, servo motor, stepper motor, etc.) working principles and usage, Robot programming with AD conversion and interfacing different hardware, sensors, etc, Control theory of robotics; Obstacle avoidance, object tracking, and motion control, etc; Advance Robotic control and operations.

SE 225 Data Communication and Computer Networking

Overview of Data Communication, Components of Data communication, Data representations, Data flow, Physical Structures of data communication, Topologies: Star, Ring, Bus, Mesh, And Hybrid Categories of Network- LAN, MAN, WAN, Internet Protocols and Standards, Data and Signals, Periodic Analog Signal, Digital Signal, Analog Data, Digital Data, Frequency, Bandwidth, Bit Rate, Transmission Impairments: Attenuation, Distortion, and Noise, Performance Measure: Throughput, Bandwidth, latency. Layered Task, Internet Model: Peer-to-Peer Process, Organization of the layer, Layers in the OSI Model. TCP/IP model, OSI Vs TCP/IP model. Physical Layer, Transmission Media, Guided media (Coaxial Cable, Twisted pair cable, Optical fiber). Unguided Media, Advantages and Disadvantages of various media, Framing, Fixed- Size Framing, Variable-Size Framing, Flow Control, Error Control Protocols, Types of Errors, Redundancy, Detection Versus Correction, Forward Error Correction Versus Retransmission. Hamming Distance, Minimum Hamming Distance, Linear Block Codes, One dimensional parity check, Two Dimensional parity checks, Cyclic Redundancy Check, Checksum, Basic concepts of Network layer, IP Address, Decimal-Binary data conversion, IP Address Classes, Network mask, Classful Address, Classless Address (CIDR), Basics of FLSM, VLSM, Sub netting concepts, VLSM for Classful Addressing, FLSM for Classless Addressing, Routing Table, Routing Protocol, Autonomous System, RIP, Transport layer Process-To-Process, Delivery, Client/Server, SCTP, TCP, UDP, Application Layer Overview, HTTP, DNS.

SE 226 Data Communication and Networking Lab

Overview of Networks and layered communications, understanding of Network equipment, wiring in detail, CAT6 UTP EIA/TIA 568A/B straight and cross-over wiring, testing, Overview of IP Addressing and sub-netting, static IP setting on Linux machine (Ubuntu) / Windows and testing, IP address and Packet Tracer, Creation of a LAN and connectivity test in the LAN, creation of VLAN and VLAN trunking, Basic concepts of Router Configuration -Static Routing Implementation, Implementation of Dynamic Routing (RIP, OSPF, BGP), Router Configuration

using CLI, Firewall Implementation, Router Access Control List (ACL), Packet capture and header analysis by Wireshark (TCP, UDP, IP), Basic Frame Relay Implementation with PVC, DNS, Web, DHCP, FTP server configuration

SE 231 System Analysis & Design Capstone Project

The "System Analysis and Design" course provides students with a comprehensive exploration of fundamental concepts in system analysis and design, emphasizing practical applications. Beginning with foundational topics such as planning, proposal development, and feature listing, the course progresses to delve into use case design, UI principles, and prototyping. Students then navigate the intricacies of databases, including RDBMS, design principles, and normalization. Subsequent weeks involve visualizing system processes through Activity and Sequence Diagrams, with a focus on implementing the MVC architecture. The course culminates in a semester-long group project, where students apply their knowledge to analyze, design, and implement a system, fostering a hands-on understanding.

SE 234 Theory of Computing

Basics of finite automata, Finite automata Identification and Computation, Deterministic Finite Automata (DFA), Non-deterministic finite automata (NFA), NFA to DFA conversion (subset construction method), Regular expressions (RE) Basics, RE to Regular language and Regular language to RE, RE to Finite Automata, DFA to RE, Context free grammar (CFG) Basics, CFG construction, Turing machine (TM).

SE 311 Design Pattern

The four pillars of object-oriented programming are abstraction, Encapsulation, Inheritance, and Polymorphism. Abstract class and interface. Definition of Refactoring and Code Smells. Different types of code smells and refactoring techniques. Use of different refactoring techniques for different purposes. Introduction to Design Patterns. Definition and classification of design patterns. Definition, Use Case, Code examples of Creational Patterns – Singleton, Factory Method, Abstract Factory, Prototype, Builder. Definition, Use Case, Code examples of Creational Patterns – Adapter Pattern, Bridge Pattern, Composite Pattern, Decorator Pattern, Facade Pattern, Flyweight Pattern, Proxy Pattern. Definition, Use Case, Code examples of Creational Patterns – Chain of Responsibility Pattern, Command Pattern, Interpreter Pattern, Iterator Pattern, Mediator Pattern, Memento Pattern, Observer Pattern, State Pattern, Strategy Pattern, Template Pattern, Visitor Pattern. Design principles (SOLID)- Single Responsibility Principle, Open Close Principle, Liskov Substitution Principle, Interface Segregation Principle, Dependency Inversion Principle.

SE 312 Software Quality Assurance & Testing

Fundamentals of Testing, testing terminology, necessities of testing, Verification, validation, Testing Principles, The Psychology of Software Testing, Testing throughout Software Life Cycle,

Software Development Model, V Model, Test Levels, Test Types, Black Box Testing, Basics on black box testing, types of black box testing, Boundary Value Analysis, Equivalence Partitioning, Decision Table based Testing, State Transition based Testing, Use Case Testing , Test Planning & Documentation, Analyze the product, Design the Test Strategy, Define the Test Objectives, Define Test Criteria, Resource Planning, Plan Test Environment, Schedule & Estimation, Determine Test Deliverables, Test Case design, Test report, White box testing, Basic Path Testing, Cyclomatic complexity, Statement coverage, Branch coverage, Condition Coverage, Path Coverage, Bug Life Cycle, Capability Maturity Model, Capability Maturity Model Integration, Quality Metrics, Mutation Testing, Decision mutations, Mutation Testing, value mutations, Statement mutations, Test tools, Types of Test Tools, Selection of test tools.

SE 313 Software Quality Assurance & Testing Lab

Fundamentals of Testing, Test Levels, Test Types, Black Box Testing, Basics on black box testing, types of black box testing, Boundary Value Analysis, Equivalence Partitioning, Decision Table based Testing, State Transition based Testing, Use Case Testing , Test Planning & Documentation, Analyze the product, Design the Test Strategy, Define the Test Objectives, Define Test Criteria, Resource Planning, Plan Test Environment, Schedule & Estimation, Determine Test Deliverables, Test Case design, Test report, White box testing, Basic Path Testing, Cyclomatic complexity, Statement coverage, Branch coverage, Condition Coverage, Path Coverage, Bug Life Cycle, Capability Maturity Model, Capability Maturity Model Integration, QualityMetrics, Mutation Testing, Decision mutations, Mutation Testing, value mutations, Statement mutations, Test tools, Types of Test Tools, Selection of test tools

GE 324 Business Analysis and Communication

Introduction to Business Analysis (BA), BA Practices and uses, Business analysis Vs Business Analytics, DIKW model, Sources of data , Understanding problem and symptom, Business analysis techniques, Requirement, SMART Technique for writing Requirements, Requirement management, Requirement Implementation, Stakeholders, Managing Stakeholders, Communication & Business Communication, Written Communication, Nonverbal communication, Office memos, face to face communication, types of report, Types of Letters, CV.

SE 321 Software Engineering Web Application

Software Engineering Web Application; History of Internet and Web; Protocols; Domain Name and URL; Attributes & Categories of Web Application; Anatomy of Web Page and URL; Overview of HTTP; Persistent and Non-Persistent HTTP; Response Time Modeling; HTTP GET & POST Request; HTTP Methods, Response Code; User-Server Interaction; Basic structure of HTML; Examples and usage of various HTML Elements; HTML Forms; Types of HTML Elements; Introduction to Bootstrap; Responsive Web Design; Rendering Mode: Flexbox, Position; CSS Variables, Web Fonts; Mobile Web; Getting started with Bootstrap; Basic Grid and Columns in Bootstrap; Using Common Components in Bootstrap; Introduction to JavaScript; Basic JavaScript Syntax; JavaScript DOM; Introduction to Server-side rendering using PHP;

Basic PHP Syntax; Introduction to Node.js & Express.js; Database Connection and CRUD operations using EXPRESS.JS; Introduction to MVC Framework; CRUD operations using Node.js & MySQL Library; Using Firebase as a Database Server; Connecting and Operating with Firebase using .

SE 322 Software Engineering Web Application Lab

Introduction to Local Web Servers (Node.js environment); Setting up the development environment; Google Developer Tools; Create a web page with Basic HTML Elements; HTML audio, video and YouTube; Develop HTML Forms; Applying different text CSS properties; Border properties and converting link to button; CSS Font properties; CSS Background property; Responsive layouts using Flexbox, media queries; JavaScript DOM manipulation, events, form validation, modal components and interactive UI; React.js fundamentals, components, JSX, props, state, events, forms, conditional rendering, list rendering and hooks; Building responsive single-page front-end applications with React; Node.js and Express.js fundamentals; Routing, Middleware, Controllers and REST APIs; Request-Response handling, req.params, req.query and req.body; MySQL integration and CRUD operations using Express; Building full-stack web applications with React, Node.js and Express.

SE 323 Software Architecture & Design

Software Architecture Introduction, Architecture Overview and Process, Architectural Structures and Views, The 4+1 View Model of Architecture, Layered Architecture, Broker Architecture, Monolithic Architecture, Microservices Architecture, Architecture Evaluation, S.O.L.I.D. Principles, Architecture Trade-off Analysis Method (ATAM) Understanding Quality Attributes, Availability, Performance, Usability, Modifiability.

SE 332 Information System Security

Information security concept, secured software development life cycle (secSDLC), secured web application design solution, threats and defense mechanisms on current web application, the classifications of malware and its prevention. Classic Cryptographic algorithm that includes Caesar Cipher, One Time Pad, Transposition, Playfair, and hill cipher. Feistel Network, Data Encryption Standard, Advanced Encryption Standard, Public Key Crypto System, RSA, Key management- Diffie-Hellman, Elliptic Curve, etc..Digital Signature, authentication protocol.

SE 342 Compiler Design

Compiler Basics, Compiler History, Language Processing System, Lexical Analysis, Syntax Analysis, Semantic Analysis, Intermediate Code Generation, Code optimization, Phases of Compiler, Grammar Constituents, Unambiguous Grammar, Left Recursion, Deterministic Grammar, Syntax Directed Translation, Parsing Techniques, LL1 Parsing, LR Parsing, Three Address Code, Operator Precedence Parsing, Code Optimization Techniques, Syntax Directed Translation Scheme.

SE 441 Software Engineering Professional Ethics

Ethics and professional ethics, ethical Frameworks: Deontological ethics (Duty-based ethics), Utilitarian ethics (Outcome-based ethics), Aristotelian ethics (Virtue ethics). Professional Ethics and responsibilities: ICT/IT/Computer ethics, some ethical guidelines for computer professionals, Examine and discuss professional codes of ethics, conduct, and practice (IEEE, ACM, SE, AITP, and so forth). Privacy and personal information: privacy, effect of ICT on privacy, Ethical and legal basis for privacy protection; Technological strategies for privacy protection. Freedom of Expression: freedom of expression and speech, hate speech, censorship, pornography, promoting an understanding of the balance between free expression and responsible communication. Intellectual Property: Intellectual property issues including copyrights, patents, and trade secrets, along with engaging in debates on software piracy, software patents, fair use, and the principles of free and open-source software. Employment and Health: impact of ICT on employment, work environments, employee monitoring, individual health and well-being, fostering an understanding of the ethical implications of technological advancements in the workplace. Cyber Laws: Information Communication Technology Act of 2006 and its amendment in 2013, Pornography Control Act 2012, Digital Security Act 2018; Copyright Act of 2000, Data Protection Act, Computer Misuse Act,

SE 411 Software Project Management & Documentation

Introduction to Software Project Management, understanding of the profession of project manager. Project Scope, Schedule and Budget Management. Quality issues in project, Resources, Risk and Communication Management, Network Diagram, Gantt Chart, Project Budget, PERT analysis, Quality tools-Six Sigma, Quality Plan & Quality documents, Procurement management Process. DevOps, Documentation Management Process, Documentation Communication, Risk Identification, Risk Communication, Finally project Communication, Stakeholder and integration of a project.

SE 333 Artificial Intelligence

Introduction to AI, Components of AI, Agents, Environments, and Fundamental Challenges of AI, Limitation of Electronic Systems, The Concept of Human Complex, Human Perception to Logical Sentence Conversion, Terminologies related to AI, and Forming the based idea of Search Algorithms, Working Principle for Informed Search Algorithms: BFS, and DFS. Working Principle of Uninformed Search Algorithms: GBFs, A Search. Introduction to Adversarial Search Algorithm & Designing Simple AI Game Logic, Forming Logical Sentences, Compound Statements using Logical Connectives, and Statement Modeling. Inference Algorithm, and Knowledge Engineering. Inference through Resolution, Bayesian Networks. Approximate Inference. Markov Assumption, Markov Chain, Markov Hidden Models, and Sensor Model. Concept of Optimization, and Formation of the State Space Landscape, AI Optimization Algorithm Design. Different Existing Optimization Algorithms, and Their Applications. Mathematical Background of Machine Learning. Learning Algorithm Formation, and Optimization. Single Layer Neural Network, Deep Neural Network, Activation Functions, Backpropagation Algorithm, Handling Overfitting, and Underfitting, Different Optimization

Algorithms. Convolutional Neural Networks. Syntax & Semantics, Formal Grammar to Context Free Grammar, Tokenization, Markov Model. Text Categorization, Naïve Bayes, Smoothing (Additive, Laplace), Topic Modeling. Word Vector Formation, Training Neural Network for NLP, Evaluating the Performance of the Trained Model, Exploring the Limitations of the NLP Project.

SE 334 Artificial Intelligence Lab

The "Artificial Intelligence Lab" course provides students with a comprehensive exploration of Basic Python Programming, Implementation of Search Algorithms, Knowledge Representation using Propositional Statements, Dealing with Uncertainty, Optimization Algorithms Implementation, Machine Learning Coding, Implementation of Neural Network, and Natural Language Processing.

SE 544 Introduction to Machine Learning

Definitions, Types, K-Nearest Neighbor, Euclidean Distance, Manhattan Distance, Differentiation, Partial Differentiation, Gradient Descent, Hypothesis Function, Loss Function, Learning Rate, Learning Rate Scheduling, Early Stopping, Stochastic Gradient Descent, Batch Gradient Descent, Mini-Batch Gradient Descent, Simple Linear Regression, Multiple Linear Regression, Logistic Regression, Binary Log-Loss Function, Polynomial Regression, Overfitting And Underfitting, Bias-Variance Trade Offs, Regularization Methods, Ridge Regression, Lasso Regression, Elastic Net, Ensemble Learning: Voting, Bagging, Pasting, Random Patches, Random Subspaces, Random Forest, Boosting, and Stacking, Model Selection: Hyperparameter Tuning, Grid Search, Random Search and Cross-Validation, Model Evaluation: Bootstrapping, Mean Absolute Error, Mean Square Error, Root Mean Square Error, Confusion Matrix, Type I Error, Type II Error, Accuracy, Precision, Recall, Sensitivity, Specificity, True Positive Rate, False Positive Rate, AUC-ROC, Precision-Recall Curve, Feature Engineering: Normalization, Standardization, and Augmentation, K-Means Clustering, Agglomerative Clustering, Single Linkage, Complete Linkage, Average Linkage, DBSCAN, Dimensionality Reduction: PCA and t-SNE.

SE 331 Software Engineering Design Capstone Project

The “Software Engineering Design Capstone Project” course provides hands-on experience and practical application of web and software development concepts using web technologies. Students will be guided through a systematic approach to develop a project, emphasizing real life problem-solving skills and logical thinking. The course covers a range of topics from problem formulation to the evaluation of solutions, ensuring that students acquire a comprehensive understanding of the software and web development process. The course encompasses front end web technologies, framework and backend web technologies and frameworks. These core concepts form the backbone of the curriculum, providing students with a comprehensive understanding of essential web application principles.

EMP 101 Employability 360

Overview - Employability 360, self-assessment, career mapping and planning, transferable soft skills, administrative tools, business communication tools, collaboration and teamwork, interview skills, finding cutting edge engineering technology and market research, engineering tech skills showcasing for Software/Computer Engineering students, Foster adaptability & lifelong learning in today's rapidly changing tech-job market.

SE 444 Data Warehouse and Data Mining

Transactional Processing Vs. Decision Support Systems, Data Warehousing Fundamentals, Characteristics of Data Warehouse, Data Warehouse Architectures, Maturity Models, Potential Applications, OLTP, OLAP, Data Cubes, Fact Tables, Dimensions, Relational Models, Integrity Rules, Entity Relationship Diagrams, Modification Anomalies, Database Normalizations, Star Schema, Snowflake Schema, Constellation Schema, Time Representation, Types of Change, ETL, Naïve Bayes, Train-Test-Validation Set, Cross-Validation, Bootstrapping, Mean Square Error, Mean Absolute Error, Root Mean Square Error, Confusion Matrix, Accuracy, Precision, Recall, F1 Score, Association Rule Mining, Apriori Algorithm, Linear Regression, Logistic Regression, Decision Tree, K-Means Clustering, Agglomerative Clustering, Graph Theory, Graph Clustering.. Community Detection, Girvan Algorithm, Louvain Algorithm.

SE 447 Human Computer Interaction

Foundations of Human Computer Interaction (Human Capabilities, Computer usability Interaction Paradigms), The Design Process (Interaction Design Basics, HCI in the Software Process, Design Rules, Universal Design), Implementation Support (Implementation Tools), Device Evaluation and User Support (Evaluation ,User Support), User-Centered Design (Participant Observation and User Personas, Low-Fidelity Prototyping, Usability Testing Computer Prototyping), Users Models (Cognitive Models, Socio-organizational Issues and Stakeholder Requirements), Task Models and Dialogs (Analyzing Tasks, Dialog Notations and Design), Groupware, Ubiquitous Computing, Virtual and Augmented Reality, Hypertext and Multimedia (Groupware and Computer-supported Collaborative Work, Ubiquitous Computing, Virtual Reality and Augmented Reality, Hypertext, Multimedia and the World Wide Web), Challenges and technologies (Perception and Visualization, Design and evaluation challenges), Interactions (Wearable and Mobile Interaction, UbiComp, Haptics), Physiological & Social Computing (Tangible and Gestural interfaces, Physiological Computing, Social Computing, Digital addiction), Behavioral Computing (Human persistency and actuator compliance).

SE 599 Research Methodology & Scientific Writing

The aim of this course is to develop students' knowledge and understanding of the role and conduct of quantitative and qualitative research methods in planning [and urban design]. Intellectual and methodological debates will be discussed in order to assist students to develop informed opinions and a critical appreciation for other's research. The imperative for ethical research practice will be presented. The course equips students with the skills to review and

conduct methodologically sound research as a part of their professional work. Students develop the skills to recognise and reflect on the strengths and limitations of different research methodologies, understand the links between theory and practice, critically assess research, and address ethical and practical issues. The course takes a step-by-step approach to the design and implementation of quantitative and qualitative techniques including case study and precedent studies, surveys, interviews, focus groups, participant observation, textual and media analysis. How to manage and analyse data (including computer assisted), and how to write up and present findings are core components of this course. Students will be equipped with the knowledge and ability to undertake original research projects and develop a set of transferable workplace skills.

SE 442 Management Information System

Management of Information Systems addresses the management of the information resources from a senior management viewpoint. This course provides students with an overview of the utilization of business application software and problem-solving using that software. The course provides instruction so students can use information technology to create collaborative environments for employees, customers, and suppliers; securely manage information assets including data and Internet-based technologies; manage system applications and data to best support business decisions operational excellence through ERP, SCM and CRM; and understand the fundamentals of how to build and manage local and global information systems. Further, this course will provide hands-on experience with data visualization, predictive analytics, prototype development and SQL data management.

RE 331 Embedded Programming

Introduction to Embedded Programming, Low-Level Programming for Embedded Systems, Real-Time Operating Systems (RTOS), Device Drivers and Hardware Interfacing, Debugging and Testing Techniques for Embedded Systems, Power Optimization Techniques in Embedded Programming, Communication Protocols for Embedded Systems, Embedded System Security, ROS, Important ROS Concepts, ROS Commands, Middleware and Simulation, Advanced ROS Concepts, Project Development, Compilers for Embedded systems, Dynamic voltage scaling, Dynamic power management, Interrupts, timers, and advanced features in microcontroller programming, Advanced motor control techniques, Overview of robotics middleware (e.g., Robot Middleware (RTM), DDS), Understanding control systems in robotics PID controllers and their applications, Trajectory planning and motion control, Hands-on projects on robot control systems, Sensors and actuators for robotics applications, Circuit design and analysis, prototyping circuits on breadboards and PCB's, Overview of Arduino platform, Arduino IDE and basic programming, GPIO programming with Arduino

RE 332 Embedded Programming Lab

Introduction to Arduino and ESP Microcontrollers, Programming with Arduino and ESP, Real-time Operating Systems (RTOS) Implementation, Hardware Description Languages (HDLs) and FPGA Programming, IoT Concepts and Applications with Arduino and ESP, Wireless

Communication Protocols in Embedded Systems, Building a Wireless Sensor Network, ROS Introduction, ROS Setup, ROS setup and run, Integrating Arduino and ESP with ROS, Project Development.

RE 411 Embedded System Design and Development

Overview of Embedded Systems, Importance of Embedded Systems in various applications, Overview of robotics applications and industries, Introduction to Microcontrollers and Microprocessors, types, Understanding electronic components used in robotics, Sensors and actuators for robotics applications, Circuit design and analysis, prototyping circuits on breadboards and PCB's, Overview of Arduino platform, Arduino IDE and basic programming, GPIO programming with Arduino, Sensor interfacing with Arduino and ESP boards, analog electronics, digital electronics, tri-state outputs and logic gates, VLSI and Integrated circuit design, Electronic design Automation tools, Embedded Firmware, Understanding the kinematics of robotic systems, Task level concurrency management, High level optimizations (Loop tiling/blocking, loop splitting, Array folding), Compilers for Embedded systems, Dynamic voltage scaling, Dynamic power management, Interrupts, timers, and advanced features in microcontroller programming, Advanced motor control techniques, Overview of robotics middleware (e.g., Robot Middleware (RTM), DDS), Understanding control systems in robotics PID controllers and their applications, Trajectory planning and motion control, Hands-on projects on robot control systems.

RE 412 Embedded System Design and Development Lab

Microcontroller Architecture and I/O Interfacing, Advanced Programming with Arduino and ESP, Embedded Programming Techniques, Actuator Control and Closed-Loop Systems, Implementing Navigation in ROS Projects, Advanced ROS Features and Applications, Security Measures in Embedded Systems, Optimizing Code for Performance, Project-Based Experiments and Project Development.

RE 421 Robotic Process Automation Design & Development

This course includes history of Automation; Story of Work; Introduction to RPA; RPA vs Automation; RPA and AI; RPA and emerging ecosystem; Industries best suited for RPA; Processes that can be automated, UiPath and its Products; Robots and their Types; Studio Overview; Orchestrator; UiPath Studio Installation & Updating; The User Interface; Features of Studio; Building 'Hello World' Robot, Variables and their Types; Variables Panel; Scope of Variable; Arguments; Arguments Panel; Argument Directions; Arguments vs. Variables, UI interactions; Input Actions and Input Methods; Containers; Recording and its types; Selectors and their types; Anchors; Debugging Selectors, Sequences; Control Flow and Its Types; Decision Control; Loops; Other Control Flow Activities; Flowcharts; Error Handling, Data Manipulation and Its importance; String Manipulations ; Data Table Manipulations; Collection, Types and Manipulations, Extraction and Its Techniques; Automation Techniques; Orchestrator Overview; Publishing a Robot to Orchestrator; Orchestrator Functionalities.

RE 422 Robotic Process Automation Design & Development Lab

Robotics Process Automation (RPA) Lab course introduces students to the fundamental principles and practical application of RPA using UiPath Studio. Beginning with an overview of automation history and the RPA landscape, students delve into UiPath Studio installation, user interface navigation, and the construction of foundational robots. Subsequent weeks cover essential topics such as UI interactions, control flow, error handling, data manipulation, and automation techniques. Advanced concepts, including the integration of RPA with AI, are explored, followed by an examination of industry-specific applications and case studies. The course culminates with a focus on Orchestrator functionality, exploration of emerging trends in RPA, and hands-on preparation and presentation of a final project, ensuring students acquire comprehensive skills in designing, implementing, and managing RPA solutions for real-world scenarios

CS 211 Cyber Security Fundamentals

Cybersecurity is one of the most significant challenges of the contemporary world, due to both the complexity of information systems and the business they support. Software running on current systems is exploited by attackers despite many deployed defense mechanisms and best practices for developing new software. In this course students will learn about overview of cybersecurity domains, global challenges related to cyber security, cyber security governance, concepts of digital trusts, cyber risk, common attack types and attack vectors, implementing cyber security controls, security architecture principles, different architect model like OSI, defense in depth, information flow control, isolation and segmentation, logging, monitoring and detection, encryption fundamentals, techniques and applications of encryption, security of networks, security of systems, security of applications, security of data, risk assessments, vulnerability management, penetration testing, network security, operating system security, application security, data security, security implications and adoption of evolving technology, like threat landscape, cloud and digital collaboration, block chain, zero trust, privilege access management (PAM), security devices, cyber security incident management, investigations, legal holds and preservation, forensics, disaster recovery plan, business continuity plans, etc. The students will work with real-world problems and technical challenges by implementing security solutions in web applications.

CS 418 Network & Communication Security

In this course, students will learn about Basic Computer Networks, Introduction to Computer Security, Aspects of Computer Security, Cryptographic Processes, Network Security-issues, Cryptographic Hash Functions, Security Workshop, Firewall Configuration and Administration, Overview of VPN, Advanced Features of VPN, Advanced Routing, NAT & PAT, Network Monitoring and Tools and Network Security infrastructure.

CS 422 Digital Forensics

The necessity of Digital Forensic and Computer Forensics Investigation Process, Understanding the Hard Disks, Understanding the File Systems, Data Acquisition and Duplication, Defeating

Anti-Forensics Techniques, Fundamentals of Operating Systems Forensics, Windows Forensic and Linux Forensic, Fundamentals of Network Security Architecture, Different Network Attack Methodologies, Network Attack Forensics, Fundamentals of Web Application Security Posture, Web Application Attack Methodologies, Investigating Web Attacks, Database Forensics, Cloud Forensics, Investigating Email Crimes, Malware Forensics, Mobile Forensics, IoT Forensics, Fundamentals of Web Application Security Posture, Web Application Attack Methodologies, Investigating Web Attacks, Database Forensics, Cloud Forensics, Investigating Email Crimes, Mobile Forensics, Malware Forensics, Importance of different types, Purposes, Scope of Forensic.

DS 331 Introduction to Data Science and Data Management & Analysis

Intro to Data Science, Learn about Data Scientist, Data Analyst, and Data Engineer, Old Problems, New Problems, Data Science Solutions, Applications of Data Science, Data Science Real-Life Problems and Solutions, Data Sets: Powering Data Science, Types and Sources of Data, Data Quality and Data Collection, Data Understanding, Data Preparation: Concepts, Data Management for Data Science, ETL Process and Architecture, Introduction to Data Analysis, Introduction to Data Visualization, Data Visualization Tools and Techniques, Data Ethics, Applications of Machine Learning and Machine Learning Workflow, Introduction to Statistics, Time Series Forecasting, Introduction to Deep Learning, Basic Neural Network Concepts, Deep Learning for Text, Image, and Sequence Data, Introduction to Large Language Models, Transformer, Attention, and Embeddings, Introduction to RAG Architecture, Business Understanding in Data Science, Model Deployment and Business Value.

DS 332 Introduction to Data Science and Data Management & Analysis Lab

This course provides a broad introduction to Data Science: (i) Learn about data science engineer and data analyst, Careers in Data Science, Application of machine learning (ii) Introduction about data science tools (iii) and data set collection and data visualization (iv) data set analysis and model applyStatistical (v) statistical basics (vi) Time series data analysis and ETL.

DS 411 Statistical Data Analysis

Data Types, Data Quality, Mean, Median, Mode, Percentiles, Quantiles, Variance, Standard Deviation, Interquartile Range, Range, Correlation, Pearson Correlation, Spearman's Rank Correlation, Histogram, Density Plot, Bar Chart, Grouped Bar Chart, Stacked Bar Chart, Heatmap, Scatter Plot, Line Plot, Sampling Methods, Probability Sampling, Simple Random Sampling, Systematic Sampling, Cluster Sampling, Stratified Sampling, Non-Probability Sampling, Snowball Sampling, Quota Sampling, Convenience Sampling, Probability Theory, Counting Theory, Permutation, Combinations, Random Variables, Discrete Random Variable, Continuous Random Variable, Probability Mass Function, Probability Density Function, Cumulative Density Function, Probability Distributions, Bernoulli Distribution, Binomial Distributions, Poisson Distribution, Hypergeometric Distribution, Normal Distribution, Uniform

Distribution, Sampling Distribution, Central Limit Theorem, Confidence Intervals, Hypothesis Testing, P-Value, Type-I Error, Type-II Error, Statistical Power, Z-Test, T-Test, Two Sample T-Test, Paired T-Test, Analysis Of Variance, Chi-Square Test, F-Statistic, Simple Linear Regression, Multiple Linear Regression.

DS 412 Statistical Data Analysis Lab

Data Types, Variables, Data Structures, Functions, Conditions, Loops, Objects, Exploratory Data Analysis, Histograms, Density Plots, Boxplots, Scatter Plots, Bar Charts, Heatmaps, Linear Regression, Logistic Regression, Independent T-Test, Dependent T-Test, ANOVA, Factorial ANOVA, Repeated ANOVA, Mixed ANOVA, ANCOVA, MANOVA, Chi-Square Test, A/B Test, Wilcoxon, Kruskal-Wallis, Friedman's Test, Multilevel Linear Model, Sample Size Calculation, Power Calculation, Intraclass Correlation, Effect Size, Report Writing.

DS 421 Machine Learning Driven Data Analysis I

Gradient Descent, Hypothesis Function, Loss Function, Learning Rate, Learning Rate Scheduling, Early Stopping, Stochastic Gradient Descent, Batch Gradient Descent, Mini-Batch Gradient Descent, Simple Linear Regression, Multiple Linear Regression, Vectorization of Gradient Descent, Logistic Regression, Binary Log-Loss Function, Polynomial Regression, Overfitting And Underfitting, Bias-Variance Tradeoffs, Regularization Methods, L1 Penalty, L2 Penalty, Ridge Regression, Lasso Regression, Elastic Net, Artificial Neural Networks, Activation Functions: ReLU, Hyperbolic Tangent Function, SoftMax, etc., Dropout Layer, Advanced Optimization Techniques: Momentum, RMSProp, and ADAM, Decision Tree, Gini, Entropy, CART Algorithm, Ensemble Learning: Voting, Bagging, Pasting, Random Patches, Random Subspaces, Random Forest, Boosting, and Stacking, XGBoost, Model Selection: Hyperparameter Tuning, Grid Search, Random Search and Cross-Validation, Model Evaluation: Bootstrapping, Mean Absolute Error, Mean Square Error, Root Mean Square Error, Confusion Matrix, Type I Error, Type II Error, Accuracy, Precision, Recall, Sensitivity, Specificity, True Positive Rate, False Positive Rate, AUC-ROC, Precision-Recall Curve, Feature Engineering: Normalization, Standardization, and Augmentation, K-Means Clustering, Agglomerative Clustering, Single Linkage, Complete Linkage, Average Linkage, DBSCAN, Anomaly Detection, Recommendation Systems, Collaborative Filtering, Content-Based Filtering, Dimensionality Reduction, Principal Component Analysis, Locally Linear Embedding, t-SNE, Reinforcement Learning, Q-Learning.

DS 422 Machine Learning Driven Data Analysis Lab

Python, its Variables and Built-in Data Structures, Object-Oriented Python, Scikit-Learn, Pandas, Matplotlib, NumPy, Supervised Machine Learning, Linear Regression, Logistic Regression, Lasso Regression, Ridge Regression, Elastic Net, Decision Tree, CART Algorithm, Support Vector Machine (SVM), Kernel Methods, Unsupervised Machine Learning, K-Means Clustering, DBSCAN, Hierarchical Clustering, Gaussian Mixture Models, Bayesian Gaussian Mixture Models, Dimensionality Reduction, Principal Component Analysis (PCA), Kernel PCA Locally Linear Embedding,

SE 431 Numerical Analysis

In this course students will be introduced to the mathematical analysis of numerical methods by emphasizing on different algorithms that are encountered in many disciplines like physical sciences and engineering. By the end of this course, students will be competent to solve complex mathematical problems using simple arithmetic operations such as bisection method, Newton Raphson's method, differentiation, matrix problems, finding roots of equations, overview of Gaussian elimination, partial pivoting, LU decomposition, boundary value.

RE 423 Advanced Robotics

The basic terminologies related to Advanced Robotics applications, advanced sensors, actuators and algorithms. Working principles of advanced robotics components like, Lidar sensor, IMU sensor, encoder sensor, GPS sensor, Depth Camera, encoder motor, hub motor, Brush less motor. In-depth knowledge about robot kinematics. Mathematical representation of forward kinematics and inverse kinematics. Introduction to Robot Operating System (ROS). Components of ROS. Brief discussion about Gazebo, RVIZ and Rqt graph. Introduction to robot navigation system. In-depth knowledge about robot localization, sensor fusion, mapping, and motion planning. Designing advanced robotics solution architecture.

RE 424 Advanced Robotics Lab

The Advanced Robotics Lab course encompasses a diverse range of topics aimed at providing students with comprehensive knowledge and practical skills in robotics. Beginning with the study and implementation of DC encoder motors and HUB motors, participants delve into sensor technologies such as IMUs and Lidar, exploring their integration with the Robot Operating System (ROS) environment. Concepts like PID control and fuzzy logic are covered for advanced motion control and decision-making. Students learn to install and utilize ROS on Ubuntu systems, write basic nodes, and implement navigation stacks for autonomous robot movement. Practical exercises involve sensor fusion, 2D map building with Lidar, and optimizing navigation algorithms. By the end of the course, participants gain proficiency in various robotic components, algorithms, and systems integration, preparing them for complex robotics applications in research and industry.

CS 335 Ethical Hacking and Countermeasure Lab

The ethical hacking and countermeasures course with LAB intends to provide students with hands-o practice to identify information system security vulnerabilities and exploit those and implement countermeasures to prevent unauthorized use of corporate information. This course will give students a practical experience of acquiring information from various sources and targeted organizations using various tools and techniques, assessing and detecting vulnerabilities, and exploiting the identified weaknesses using different ethical hacking methodologies. The tools and techniques covered in class will practically prepare students for performing hacking ethically to protect corporate information. Furthermore, it would prepare them to understand and implement appropriate countermeasures to prevent unauthorized corporate information systems.

DS 423 Machine Learning Driven Data Analysis II and Communicating Data Insights

Introduction to deep learning, create and train neural network architectures like Basic of Neural Network, Action Function, Forward Propagation, Back Propagation, Chain Model Convolutional Neural Networks, Recurrent Neural Networks, LSTMs, and Transformers, as well as how to improve them with tactics like Dropout, BatchNorm, Xavier/He initialization, and more. Students will be able to grasp theoretical principles and their industry applications using Python and TensorFlow, and to take on real-world problems like object detection, speech recognition, music synthesis, and chat bots, and more.

CS 334 Ethical Hacking and Countermeasure

In this course students will learn about Necessity of ethical hacking knowledge to protect the infrastructure, foot printing & reconnaissance process, vulnerability assessment, analyze them to prepare attack vector, exploiting vulnerabilities to hack the system, web application attack, different attack methodology, sniffing, malware analysis, cryptography & cloud computing etc.

DS 424 Machine Learning Driven Data Analysis II and Communication Data Insights Lab

This course provides a broad introduction to Data Science: (i) Learn about data science engineer and data analyst, Careers in Data Science, Application of machine learning (ii) Introduction about data science tools (iii) and data set collection and data visualization (iv) data set analysis and model apply (v) statistical basics (vi) Time series data analysis and ETL

ST 411 Agile Testing

Agile Software Development Fundamentals, Agile values and principles, Whole-team approach, Early and frequent feedback, Agile development approaches, Collaborative user story creation, Retrospectives, Continuous integration, Release and iteration planning, Differences between traditional and agile testing, Testing and development activities, Agile work products, Agile test levels, Agile configuration management, Independent testing options, Communicating test status and progress, Managing regression risk, Agile tester skills, Role of a tester in agile teams, Test-driven development, Acceptance test-driven development, Behavior-driven development, Agile test pyramid, Testing quadrants, Agile test levels and testing types, Assessing quality risks, Estimating testing effort, Acceptance criteria and adequate test coverage, Applying acceptance test-driven development, Functional black-box test design, Non-functional test design, Exploratory testing in agile, Task management tools, Communication and collaboration tools, Build and distribution tools, Configuration management tools, Test design tools, Test implementation tools, Test execution tools, Cloud-based testing tools, Virtualization tools.

ST 412 Agile Testing Lab

This course provides a practical, hands-on understanding of Agile Testing concepts, practices, and tools within an Outcome-Based Education (OBE) framework. Learners will explore agile values, principles, roles, and ceremonies, and apply testing activities across agile iterations (sprints). The course emphasizes user stories, acceptance criteria, TDD, BDD, exploratory testing, risk-based testing, and agile test metrics. Students will gain practical experience with agile testing tools, automation basics, CI/CD pipelines, and structured agile test documentation. Through continuous lab activities, tool-based exercises, sprint-level reporting, and evaluations, this course prepares learners for real-world agile QA environments

ST 421 Testing with Generative AI

Testing with Generative AI extends the Foundation Level and provides testing and quality-engineering professionals with a structured, hands-on path to master how, when, and why to use Large Language Models (LLMs) and other generative-AI techniques across the entire test process – from requirements analysis and test design to automation, reporting, and continuous improvement. It not only explains the fundamentals of GenAI but also builds practical skills through real-world prompt-engineering patterns and applied testing use cases. Testers will learn how to integrate GenAI capabilities responsibly while managing key risks such as hallucinations, bias, security, privacy, and environmental impact.

ST 422 Testing with Generative AI Lab

This laboratory course provides hands-on experience in applying Generative AI and Agentic AI concepts to real-world software testing activities. Students will use LLM-powered tools to support test analysis, test design, test automation, defect reporting, and test management. The lab emphasizes AI-assisted testing workflows, prompt engineering for testing tasks, autonomous test agents, and integration with modern automation and CI/CD tools.

ST 413 Test Automation & Test Management

This course covers the principles, tools, and practices of modern test automation and test management. Students will learn the fundamentals of automation, including automation scope, framework architecture, locator strategies, wait mechanisms, assertions, and the use of industry-standard tools such as Selenium, Cypress, Postman, and JMeter. The course also introduces API test automation, performance testing, and Behavior-Driven Development (BDD) through Cucumber or similar tools. In the second part, the course focuses on core test management activities. Topics include test planning, test strategy development, estimation techniques, test documentation (test cases, scenarios, RTM, summary report), and defect management using JIRA. Students explore test monitoring and control, quality metrics, regression management, and reporting practices used by QA teams. The course concludes with integrating automation into CI/CD pipelines, emphasizing continuous testing, version control, pipeline execution, and automation maintenance. By the end of the course, learners will be able to design

automation scripts, build automation frameworks, and manage complete testing processes aligned with industry standards.

ST 414 Test Automation & Test Management Lab

This lab course provides hands-on experience in test automation and test management practices used in modern software development. Students will work with automation tools such as Selenium, API testing tools, and CI/CD pipelines while also learning test planning, estimation, documentation, defect management, and test metrics. The lab emphasizes real-life project scenarios, automation framework development, test execution, reporting, and integration with DevOps workflows to ensure industry readiness.

ST 423 Software Performance Engineering & Security Testing

This course integrates Software Performance Engineering and IT Audit & Software Quality Assurance, providing students with a strong foundation in evaluating system performance, governance, and controls. Topics include performance testing fundamentals, workload modeling, SLAs/SLOs/KPIs, performance risks, test planning, monitoring, bottleneck analysis, and the impact of system architecture and environments on performance. Students will learn to interpret performance data, conduct root-cause analysis, and design effective performance test strategies. The IT audit component introduces software quality concepts, IEEE standards, verification and validation, IT and data governance, lifecycle frameworks such as IEEE 12207, and the role of audit checkpoints in SDLC. Additional areas include test design, estimation, defect and incident management, risk and vulnerability assessment, information security frameworks (ISO, COBIT, OCTAVE), and the conceptual application of IT audit processes, controls, and evidence collection. The course prepares learners to understand performance fundamentals, analyze performance issues, design audit frameworks, and apply audit procedures within software environments.

ST 424 Software Performance Engineering & Security Testing Lab

This course provides an in-depth, hands-on understanding of software performance testing and IT audit practices. Learners will acquire both technical and analytical skills by designing performance test plans, executing workload models, analyzing system bottlenecks, and preparing performance reports. Alongside performance engineering, students will explore IT governance, audit methodologies, security controls, risk assessment, and audit documentation techniques aligned with international frameworks such as ISO and COBIT. Through a blend of lab activities, projects, root-cause analysis, and structured audit documentation, this course prepares learners to contribute effectively to industry-level QA and IT audit environments.

GED S237 Entrepreneurship in IT Business

Entrepreneurial mindset and traits, Pre-assignment review, Finding business ideas, Considering options, Outlining and assessing ideas, Evaluating capacity for risk, Developing product ideas,

Making business connections, Identifying target market and customers, Developing value proposition, Creating financial projections, Types of business ownership, Purchasing a franchise, Purchasing an existing business, Business plan preparation, Executive summary, Pitch deck, 30-second pitch, Practicing the pitch, White papers, Basics of financial statements, Brainstorming funding solutions, Funding options, Product development steps, Protecting intellectual property, Introduction to marketing, Consumer and market analysis, Competition and self-analysis, Distribution channel analysis, Creating marketing plan, Implementing, evaluating, reviewing and revising marketing strategy, Leveraging social media, Building sales force, Branding fundamentals, Brand name development, Slogan development, Logo design, Creative branding, Office setup and location selection, Pros and cons of home-based businesses, Business launch strategies, Organizational chart setup, Recruitment and team engagement, Business operations and tracking, Business growth strategies, Entrepreneurial leadership perspectives, Case studies on entrepreneurship, Sustaining and scaling IT ventures

SE 345 AI System & Application Development

Mobile application ecosystem and evolution; mobile operating systems and framework comparisons (native, hybrid, and cross-platform); fundamentals of Dart programming language including syntax, data types, operators, control statements, functions, and object-oriented principles; class and object concepts applied in mobile app logic; introduction to Flutter framework, project architecture, and widget tree; stateless and stateful widget design; form handling and navigation; asynchronous programming and API communication; local storage mechanisms including shared preferences and SQLite CRUD operations; performance optimization, maps and geolocation, deployment processes, and professional practices in mobile app development with ethical and user experience considerations.

SE 346 AI Systems & Application Development Lab

Hands-on exercises on Dart programming, Flutter basics, OOP implementation, and widget manipulation; layout design and UI development; event handling and state management; navigation and validation; backend API integration; local database (SQLite) and CRUD operations; app deployment practice.

SE 344 Distributed Systems and Cloud Computing

Concepts of Distributed Computing, fundamentals of Cloud concepts and capabilities across the various Cloud service models including Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS), Introduction to AWS Cloud, AWS Global Infrastructure Overview, the Cloud security model, associated challenges, and implementation, security and privacy issues in cloud computing.

GEDS 416 AI for Strategic Decision Making

This course introduces students to the role of Artificial Intelligence (AI) in strategic business and organizational decision making. It focuses on how data-driven and AI-powered methods, including predictive analytics, machine learning, and decision-support systems, can be used to analyze complex business problems and guide strategic choices. The course begins with an overview of strategic decision-making frameworks and the evolving role of AI in organizations. Over the course of the subsequent units, it covers data-driven decision models, predictive and prescriptive analytics, machine learning for forecasting and classification, AI-based recommendation and optimization systems, and the ethical, organizational, and risk considerations of deploying AI in strategic contexts

SE 343 Software Maintenance

Software maintenance types, models, and lifecycle integration; Lehman's laws of software evolution and change phases; change management and impact analysis techniques; version control strategies (Gitflow, trunk-based) and configuration management with build automation; CI/CD pipelines and regression testing; software quality metrics (LOC, cyclomatic complexity, cohesion, coupling) and static analysis tooling (SonarQube); code smells taxonomy and anti-pattern detection; refactoring fundamentals and catalog (Extract Method, Move Method, Rename); structural design patterns (Strategy, Observer, Factory Method, Decorator) in maintenance contexts; advanced testing strategies including coverage criteria and performance profiling; dynamic analysis and code pruning; release management and deployment strategies; technical debt measurement and management in agile environments.

SE 431/RE 431/CS 439/DS 431 Final Year Project/Thesis/Internship Robotics &

As part of their fourth year, students are required to carry out a project and submit a technical report. This project is a substantial piece of work that will require creative activity and original thinking. Individual students are supervised while working on a project accounting for six credit hours, extending over a period of two semesters. The project aims to provide students with a transitional experience from the academic world to the professional world. It is designed to serve as a platform in which Software Engineering students engage in a meaningful design experience requiring the solution of complex engineering design projects.